

GRAPH NEURAL NETWORKS



COLD OPEN

Can we trust this person?



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

COLD OPEN

Can we trust this person?



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FORWARD · WORLD CUP WINNER

On his own, nothing looks suspicious.

COLD OPEN

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*



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*ordered to
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KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

*told how to
referee*

THE REFEREE



On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

THE REFEREE



*told how to
referee*

*stared down
through the screen*



YOU, THE VIEWER

On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

THE REFEREE



*told how to
referee*

*stared down
through the screen*



YOU, THE VIEWER

Wait... why is everyone taking orders from this guy?

COLD OPEN

Can we trust this person?



INFERRED FROM HIS CONNECTIONS

DICTATOR?

(OF THE PENALTY BOX)

COLD OPEN

Can we trust this person?



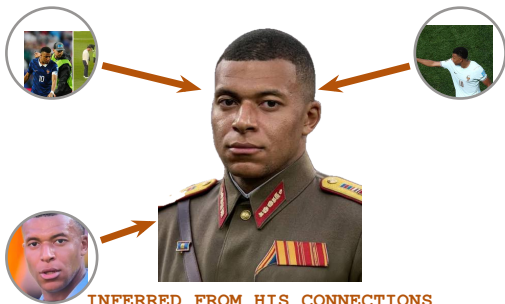
INFERRED FROM HIS CONNECTIONS

DICTATOR?

(OF THE PENALTY BOX)

What if the person itself
isn't the information?

Can we trust this person?



INFERRED FROM HIS CONNECTIONS

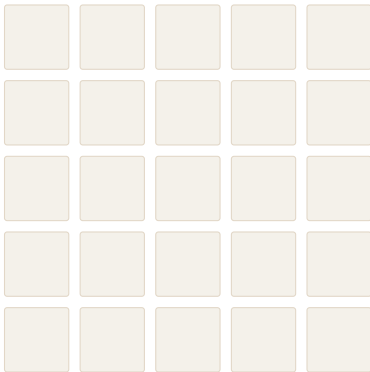
DICTATOR?

(OF THE PENALTY BOX)

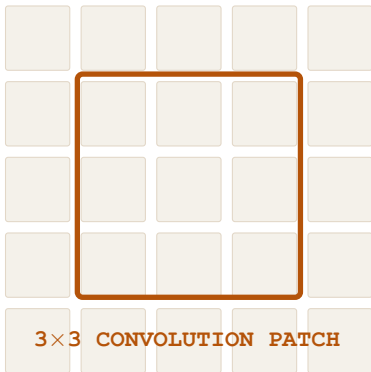
What if the person itself
isn't the information?

What if the
connections are?

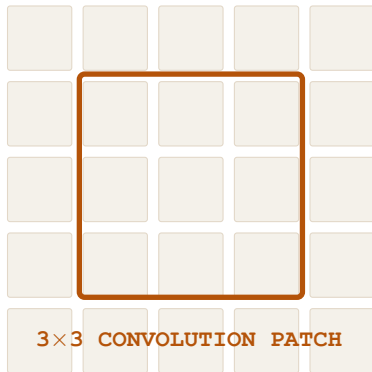
THE CNN EXAMPLE



THE CNN EXAMPLE

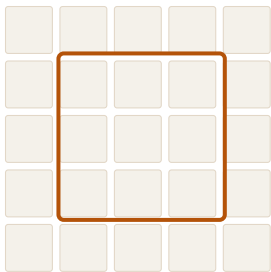


THE CNN EXAMPLE



LOCAL CONNECTIVITY · FIXED SPATIAL GRID

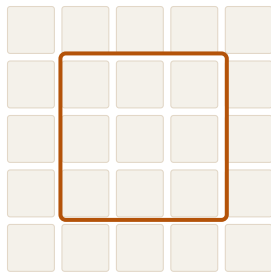
THE CNN EXAMPLE



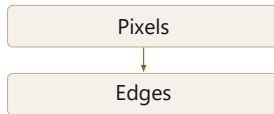
LOCAL PATCHES

Pixels

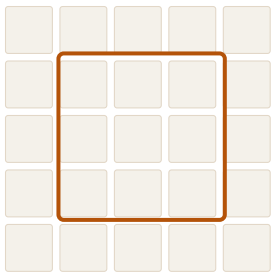
THE CNN EXAMPLE



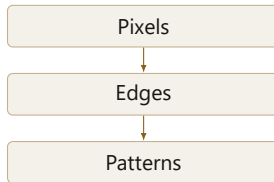
LOCAL PATCHES



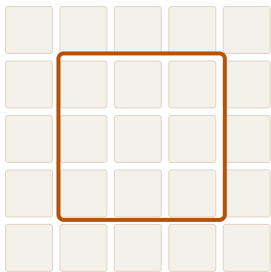
THE CNN EXAMPLE



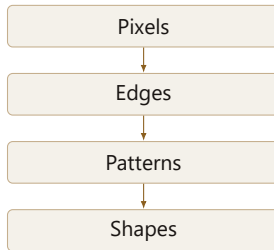
LOCAL PATCHES



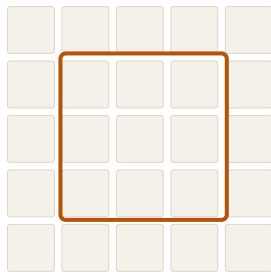
THE CNN EXAMPLE



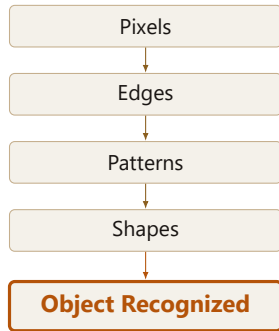
LOCAL PATCHES



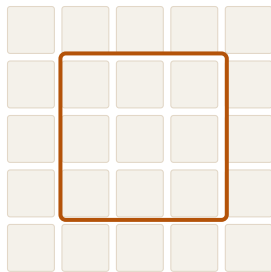
THE CNN EXAMPLE



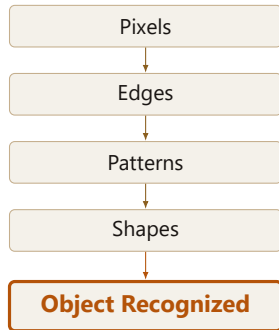
REGULAR PIXEL GRID



THE CNN EXAMPLE



REGULAR PIXEL GRID



RIGID GEOMETRIC HIERARCHY

Convolutional Neural Network

Convolutional Neural Network

CNN

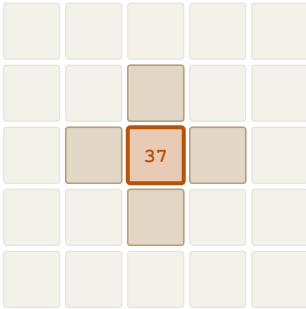
Convolutional Neural Network

CNN

BUILT FOR REGULAR EUCLIDEAN GRIDS

GRID VS. GRAPH



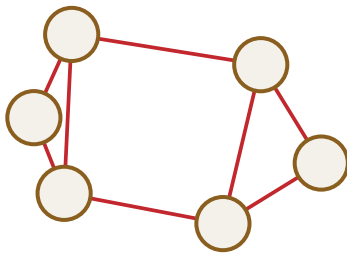


What happens when the
data has no grid?

GRID VS. GRAPH



REGULAR GRID

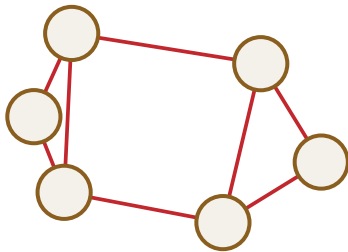


IRREGULAR GRAPH

GRID VS. GRAPH



REGULAR GRID



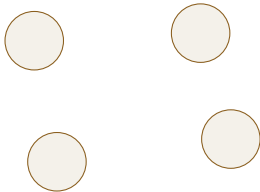
IRREGULAR GRAPH

There is no pixel 37.

NO FIXED ORDER · ARBITRARY NEIGHBORS

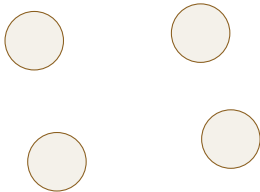
OBVIOUS SOLUTIONS

**SOLUTION 1:
DISCARD THE EDGES**



OBVIOUS SOLUTIONS

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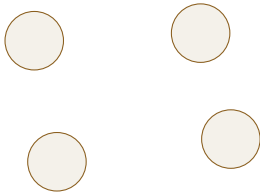


LOSES GRAPH STRUCTURE

What if we keep them?

OBVIOUS SOLUTIONS

**SOLUTION 1:
DISCARD THE EDGES**



LOSES GRAPH STRUCTURE

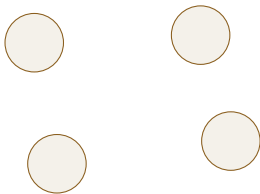
**SOLUTION 2:
FLATTEN THE ADJACENCY MATRIX**

$A \rightarrow [0, 1, 0, 1, 1, 0, 1, 0, \dots]$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

OBVIOUS SOLUTIONS

SOLUTION 1: DISCARD THE EDGES



LOSES GRAPH STRUCTURE

SOLUTION 2: FLATTEN THE ADJACENCY MATRIX

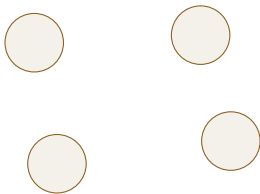
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SENSITIVE TO NODE ORDERING

OBVIOUS SOLUTIONS

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DISCARD THE EDGES



LOSES GRAPH STRUCTURE

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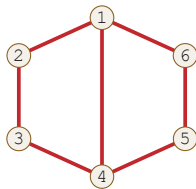
SENSITIVE TO NODE ORDERING

Why do naive approaches fail?

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

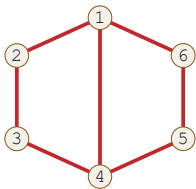
NUMBERING A



TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

NUMBERING A



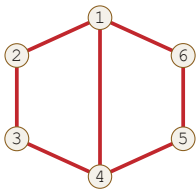
Toxicity: 0.83

ILLUSTRATIVE PREDICTION

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NUMBERING A



Toxicity: 0.83

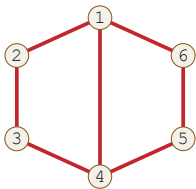
ILLUSTRATIVE PREDICTION

What if we simply
renumber the atoms?

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

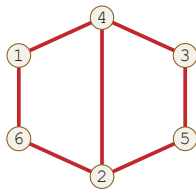
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

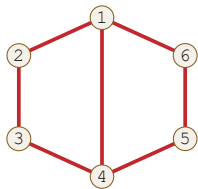
NUMBERING B



TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

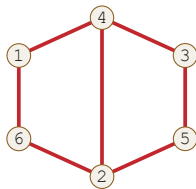
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

NUMBERING B



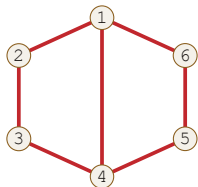
Toxicity: 0.11

ILLUSTRATIVE PREDICTION

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

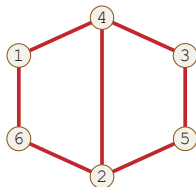
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

NUMBERING B



Toxicity: 0.11

ILLUSTRATIVE PREDICTION

PERMUTATION SENSITIVITY · ARBITRARY NUMBERING BIAS

THE FAILURE

NUMBERING A · ADJACENCY MATRIX

	1	2	3	4	5	6
1		■		■		■
2	■		■			
3		■		■		
4	■		■		■	
5				■		■
6	■				■	

Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX

	1	2	3	4	5	6
1				■		■
2				■	■	■
3				■	■	
4	■	■	■			
5		■	■			
6	■	■				

Toxicity: 0.11

SAME BONDS
DIFFERENT FILLING

THE FAILURE

NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX



Toxicity: 0.11

SAME BONDS
DIFFERENT FILLING

THE FAILURE

NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX



SAME BONDS
DIFFERENT FILLING

Toxicity: 0.11

The network learned our filing system.

THE FAILURE

NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX



Toxicity: 0.11

SAME BONDS
DIFFERENT FILLING

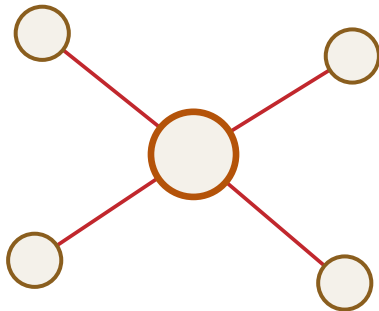
**The network learned our filing system.
It never learned the chemistry.**

What is this node?



What **is** this node?

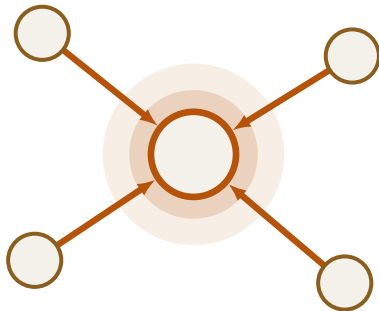
What **is** connected to it?



What **is** this node?

What is connected to it?

Who influences this node?

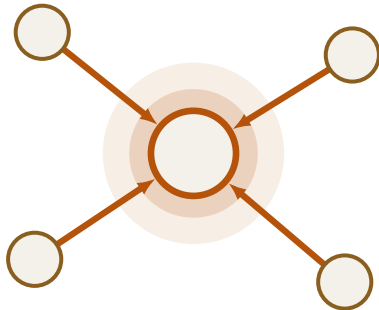


What **is** this node?

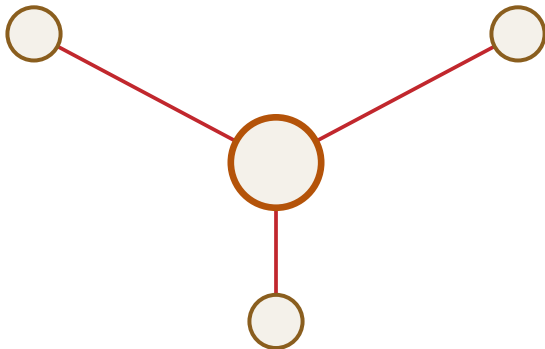
What is connected to it?

Who influences this node?

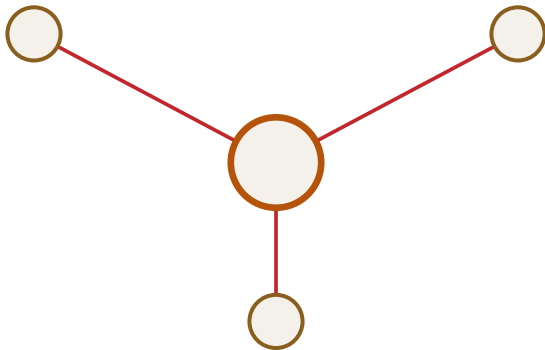
RELATIONAL REASONING



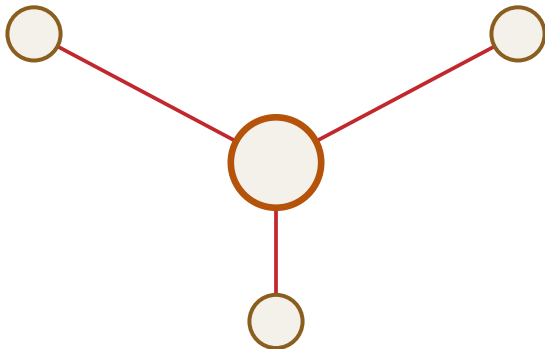
We need to use the neighbourhood.



That is the idea behind a **Graph Neural Network**.

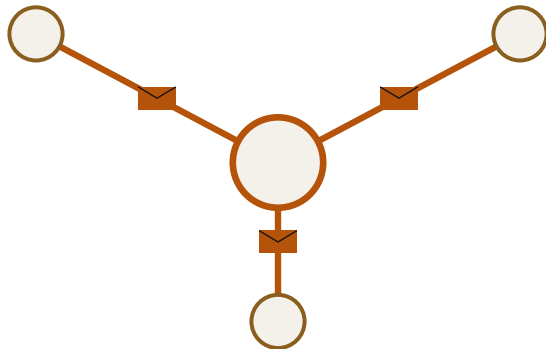


It passes messages between connected nodes.



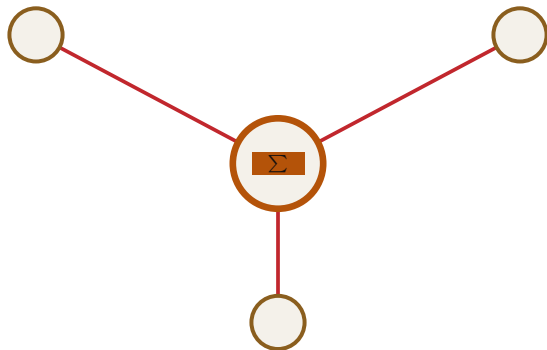
MESSAGE PASSING

It passes messages between connected nodes.



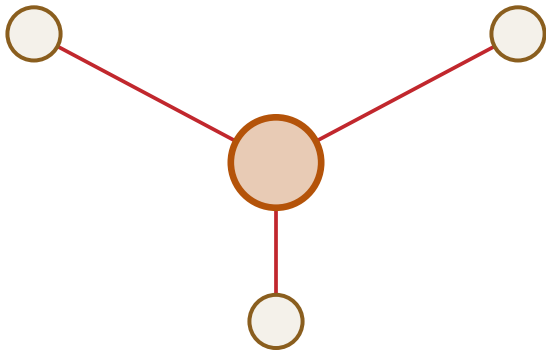
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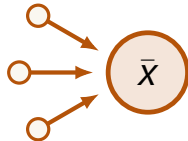
MESSAGE PASSING

It passes messages between connected nodes.



MESSAGE PASSING

What went wrong, and how a **GNN** fixes it

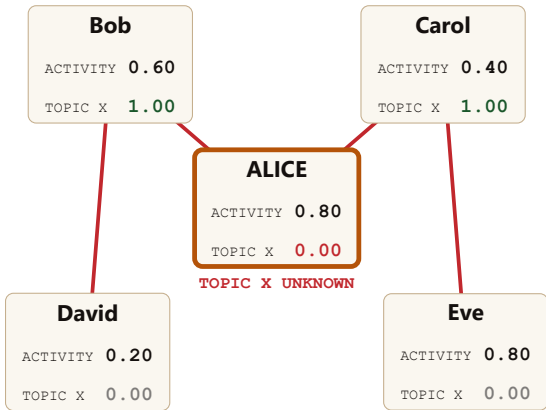


Answer depends on numbering

Order doesn't matter

Now watch five people **gossip**.

Is Alice likely to be interested in Topic X?



SLOT 1 ACTIVITY

how much they post
0 = never 1 = constantly

SLOT 2 TOPIC X

how much they engage with Topic X
0 = never 1 = all the time

THE FIVE CARDS

Bob [0.60 , 1.00]
Carol [0.40 , 1.00]
David [0.20 , 0.00]
Eve [0.80 , 0.00]
Alice [0.80 , 0.00]

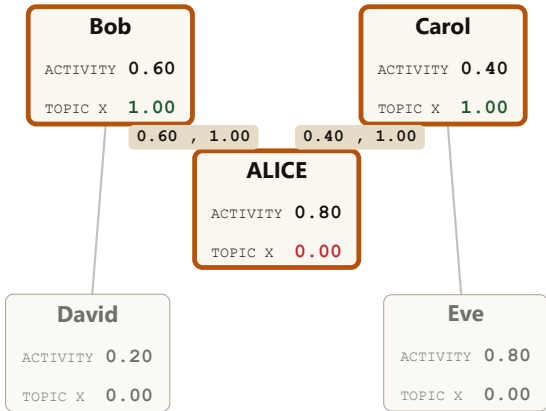
1. GATHER

2. AGGREGATE

3. UPDATE

= ONE LAYER

Step 1 of 3 — GATHER



STEP 1 GATHER

$N(\text{Alice}) = \{ \text{Bob} , \text{Carol} \}$

WHAT IS A MESSAGE?

Bob sends [0.60 , 1.00]

Carol sends [0.40 , 1.00]

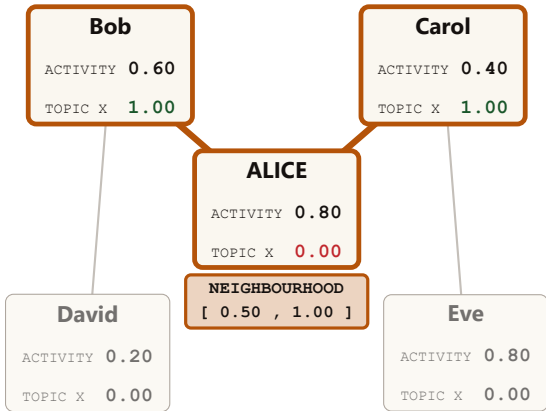
Gathering = Copying messages

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

Bob [0.60 , 1.00]

Carol [0.40 , 1.00]

Rule: average them, slot by slot.

ACTIVITY $(0.60 + 0.40) / 2 = 0.50$

TOPIC X $(1.00 + 1.00) / 2 = 1.00$

$m(\text{Alice}) = [0.50 , 1.00]$

ORDER INVARIANCE

$\text{mean}(\text{Bob}, \text{Carol}) = \text{mean}(\text{Carol}, \text{Bob})$

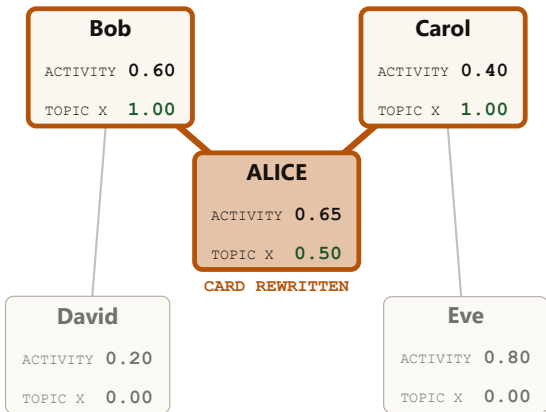
Order Invariant Aggregation

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

```
mine    [ 0.80 , 0.00 ]
theirs  [ 0.50 , 1.00 ]
```

new = 0.5 x mine + 0.5 x theirs

```
ACTIVITY  0.5(0.80) + 0.5(0.50)
           = 0.40 + 0.25 = 0.65
```

```
TOPIC X   0.5(0.00) + 0.5(1.00)
           = 0.00 + 0.50 = 0.50
```

h'(Alice) = [0.65 , 0.50]

1. GATHER

2. AGGREGATE

3. UPDATE

One blank slot, filled in by the neighbourhood.

BEFORE

ALICE	
ACTIVITY	0.80
TOPIC X	0.00



AFTER ONE LAYER

ALICE	
ACTIVITY	0.65
TOPIC X	0.50

TOPIC X: 0.00 → 0.50

READOUT LAYER

$$p = \sigma(6 \times 0.50 - 0.9) = \sigma(2.1) = 0.89$$

LIKELY INTERESTED 89%

PREDICTION DERIVED FROM CONNECTIONS

Alice was not special. Every node ran the same three steps.



ALL NODES CONVERGE TOWARD 0.50 IN SLOT 2

POTENTIAL BOTTLENECK: OVER-SMOOTHING

What the **neighbours** contribute

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

how much of them
 0.5

the shuffle-proof combiner
 average = [0.50 , 1.00]

each neighbour's current card
 Bob = [0.60 , 1.00]

my neighbours
 { Bob , Carol }

Gather · Aggregate

What **Alice** keeps, and the new card

squash it back into range how much of me $0.5 \times [0.80, 0.00]$

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

Alice's new card
[0.65 , 0.50]

Gather · **Aggregate** · **Update**

From Message Passing to Predictions



k -HOP NEIGHBORHOOD EMBEDDINGS

Fraud. Superbugs. The entire planet.

Scenario: Classifying the Dhaka Suspect



SUSPECT EMBEDDING h_{target}

Target #4091

Scenario: Classifying the Dhaka Suspect



SUSPECT EMBEDDING h_{target}

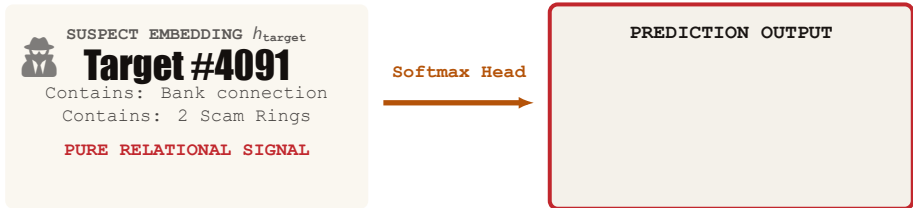
Target #4091

Contains: Bank connection

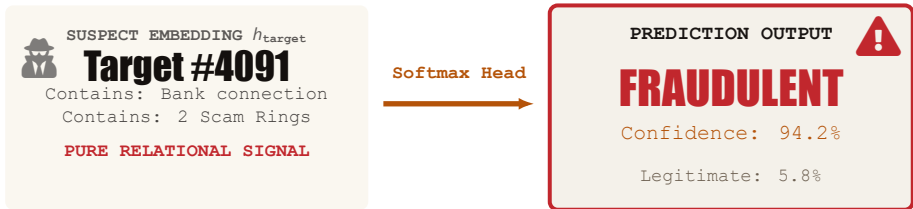
Contains: 2 Scam Rings

PURE RELATIONAL SIGNAL

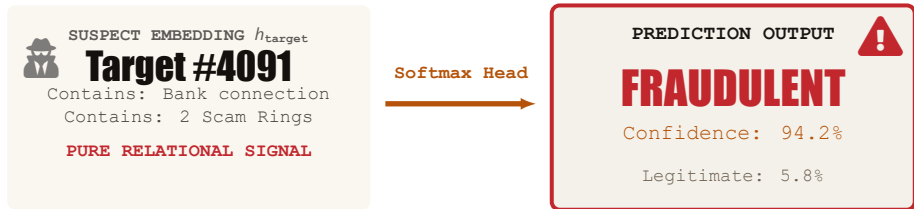
Scenario: Classifying the Dhaka Suspect



Scenario: Classifying the Dhaka Suspect



Scenario: Classifying the Dhaka Suspect



GUILT BY ASSOCIATION · LEARNED MATHEMATICALLY



NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$



NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$



EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$



NODE LEVEL

Classification

Is this account
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$$\hat{y}_v = \text{Softmax}(Wh_v)$$



EDGE LEVEL

Link Prediction

Will these two nodes
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$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$



GRAPH LEVEL

Property Test

Is this entire molecule
toxic or curative?

$$h_G = \text{Pool}(\{h_v\})$$

THE THREE GRAPH PREDICTION TASKS



NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$



EDGE LEVEL

Link Prediction

Will these two nodes
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$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$



GRAPH LEVEL

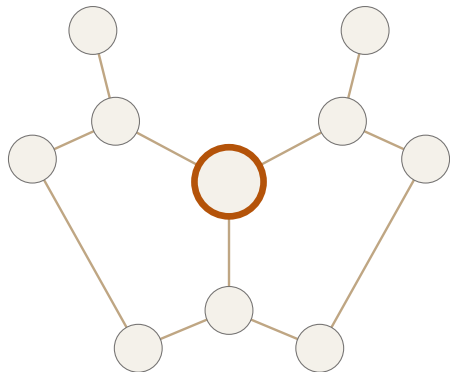
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toxic or curative?

$$h_G = \text{Pool}(\{h_v\})$$

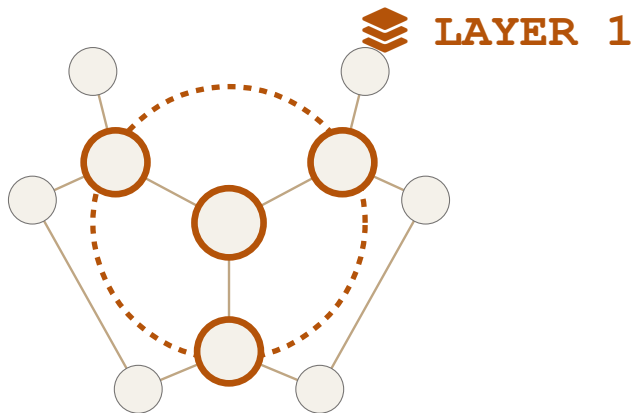
ONE MECHANISM · THREE PREDICTION SCALES

DEPTH = REACH



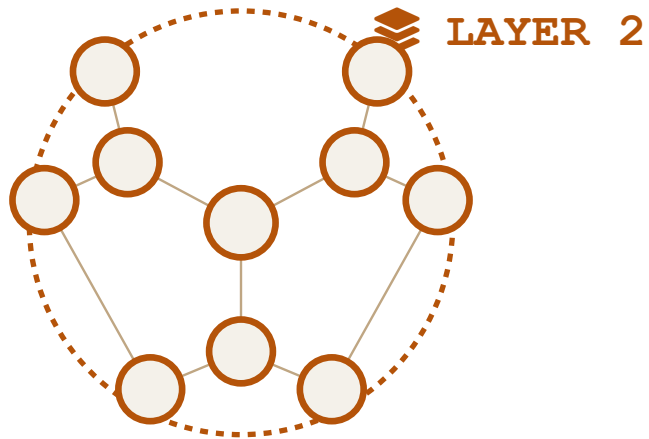
INPUT GRAPH: 1-HOP NEIGHBORHOOD

DEPTH = REACH



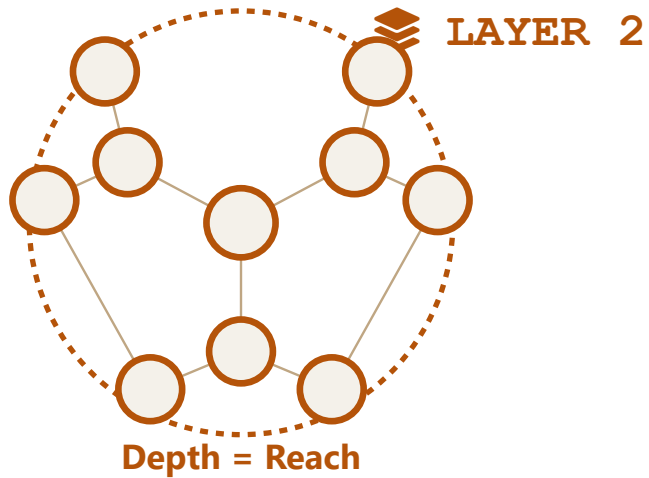
LAYER 1: 1-HOP REACH

DEPTH = REACH



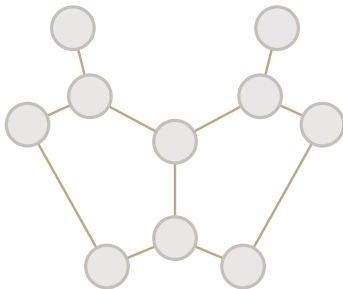
LAYER 2: 2-HOP REACH

DEPTH = REACH





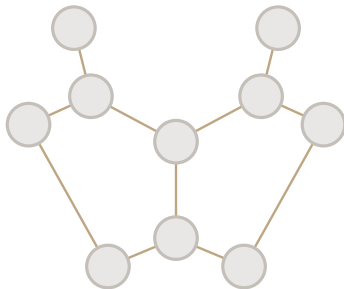
LAYER 6



OVER-SMOOTHING

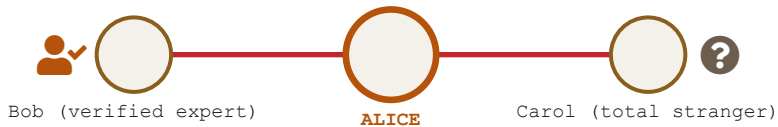


LAYER 6

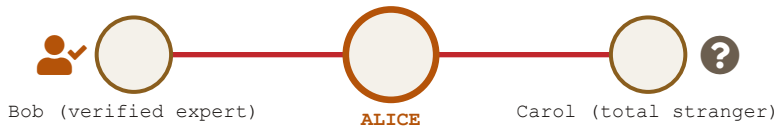


OVER-SMOOTHING
CONVERGENCE TO MEAN · LIMIT: 2-3 LAYERS

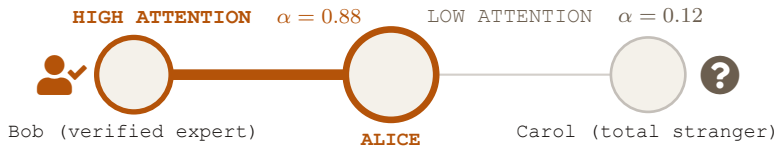
We averaged Bob and Carol equally. Should we have?



We averaged Bob and Carol equally. Should we have?

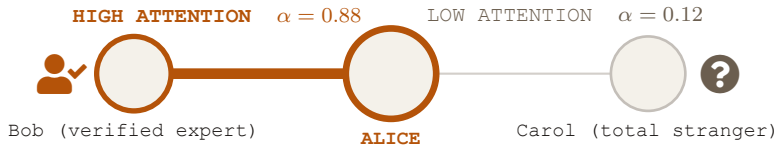


ATTENTION



LEARNED ATTENTION WEIGHTS (α_{ij})

ATTENTION



LEARNED ATTENTION WEIGHTS (α_{ij})

DYNAMIC PER-EDGE ATTENTION COEFFICIENTS



GCN

Average everyone



GCN

Average everyone



GraphSAGE

Sample a few



GCN

Average everyone



GraphSAGE

Sample a few



GAT

Learn who to trust



GCN

Average everyone



GraphSAGE

Sample a few



GAT

Learn who to trust

GATHER · AGGREGATE · ATTEND



GCN

Average everyone



GraphSAGE

Sample a few



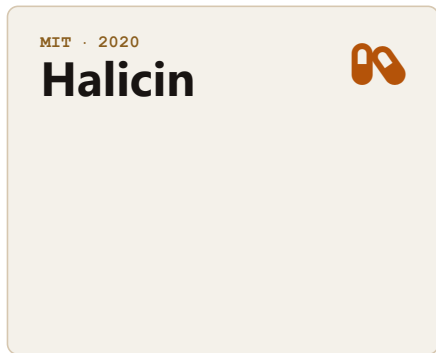
GAT

Learn who to trust

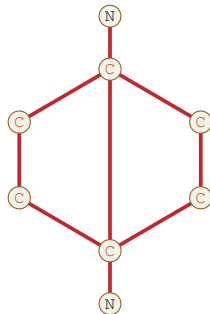
GATHER · AGGREGATE · ATTEND

HOW TO GATHER · HOW TO AGGREGATE · WHO TO LISTEN TO

CASE STUDY 1: DRUG DISCOVERY



HALICIN GRAPH REPRESENTATION



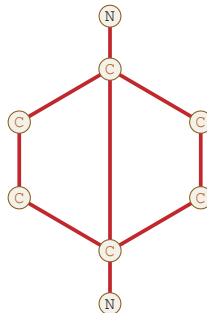
MIT · 2020

Halicin



- Screened **100 million** molecules as graphs.

HALICIN GRAPH REPRESENTATION



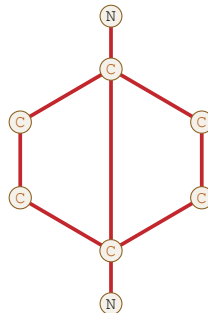
MIT · 2020

Halicin



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- Learned chemical bond topologies, not atom numbering.

HALICIN GRAPH REPRESENTATION



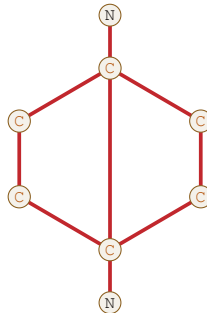
MIT · 2020

Halicin



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- Flagged a molecule humans had completely overlooked.

HALICIN GRAPH REPRESENTATION



MIT · 2020

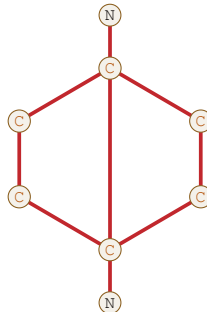
Halicin



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Killed Pan-Resistant Bacteria

HALICIN GRAPH REPRESENTATION



MIT · 2020

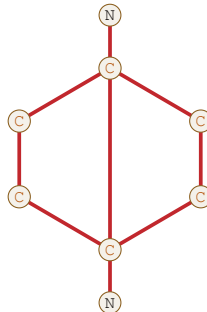
Halicin



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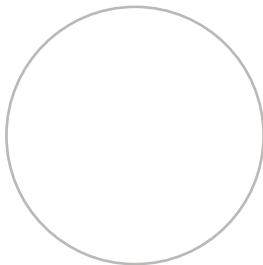
Killed Pan-Resistant Bacteria

HALICIN GRAPH REPRESENTATION



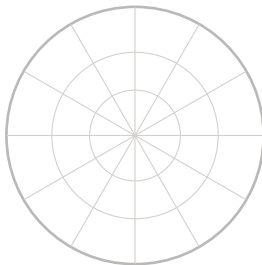


CASE FILE 02: THE ENTIRE ATMOSPHERE





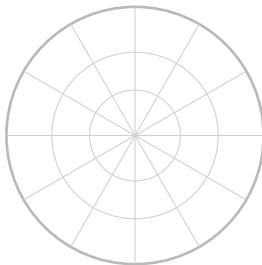
CASE FILE 02: THE ENTIRE ATMOSPHERE



ATMOSPHERIC SPHERICAL MESH GRAPH



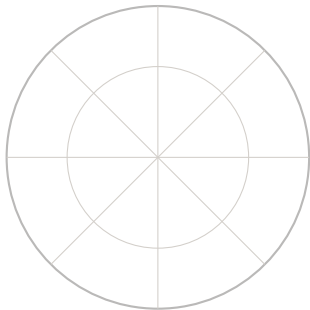
CASE FILE 02: THE ENTIRE ATMOSPHERE



ATMOSPHERIC SPHERICAL MESH GRAPH

SPHERICAL MESH GRAPH · PLANETARY AIRFLOW

CASE STUDY 2: GLOBAL WEATHER

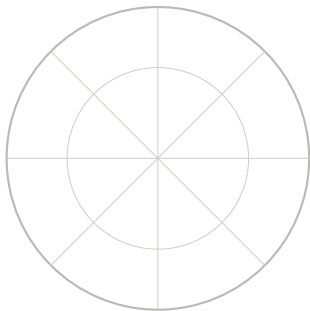


PLANETARY SCALE

DEEPMIND · 2023

GraphCast





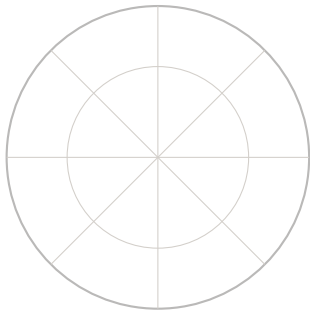
PLANETARY SCALE

DEEPMIND · 2023

GraphCast



- 10-day global forecast in **under 1 minute** on one machine.



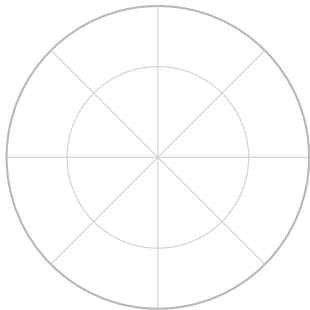
PLANETARY SCALE

DEEPMIND · 2023

GraphCast



- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on $>90\%$ of variables.



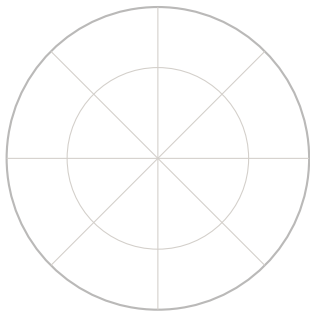
PLANETARY SCALE

DEEPMIND · 2023

GraphCast



- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on $>90\%$ of variables.
- Predicted Hurricane Lee days earlier.



PLANETARY SCALE

DEEPMIND · 2023

GraphCast

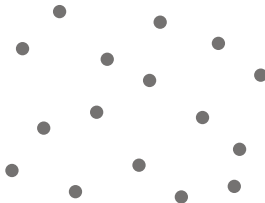


- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.
- Predicted Hurricane Lee days earlier.

Same Principle. Planetary Scale.



CASE FILE 03: CERN LARGE HADRON COLLIDER



Which hits belong to the same particle?

CERN LHC · CMS / ATLAS

Connect Dots



CERN LHC · CMS / ATLAS

Connect Dots



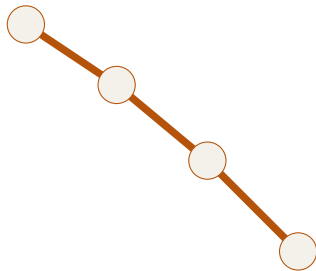
- Builds a graph of candidate hits.



CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.

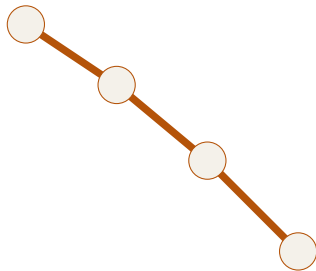


TRUE PARTICLE TRAJECTORY

CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.
- Reconstructs particle trajectories in real time.



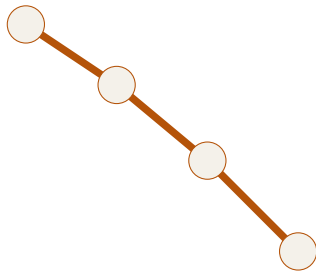
TRUE PARTICLE TRAJECTORY

CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.
- Reconstructs particle trajectories in real time.

Discovering New Physics



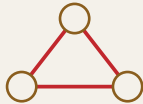
TRUE PARTICLE TRAJECTORY

THE UNIFYING PICTURE



MOLECULES

Atoms + Bonds



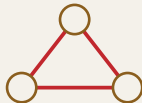
CHEMISTRY

THE UNIFYING PICTURE



MOLECULES

Atoms + Bonds

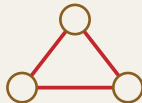


CHEMISTRY



PLANET

Grid + Airflow



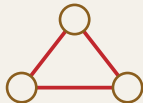
ATMOSPHERE

THE UNIFYING PICTURE



MOLECULES

Atoms + Bonds

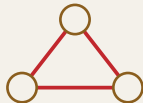


CHEMISTRY



PLANET

Grid + Airflow

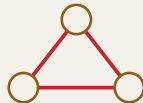


ATMOSPHERE



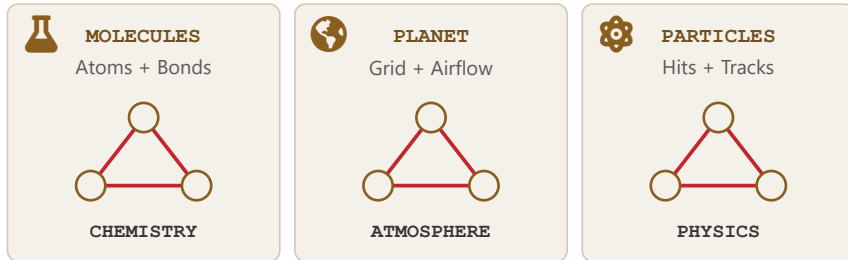
PARTICLES

Hits + Tracks



PHYSICS

THE UNIFYING PICTURE



DIVERSE DOMAINS · ONE GRAPH LANGUAGE

THE UNIFYING PICTURE

GRAPH

GRAPH

Different problems. Same underlying structure.

GRAPH

Different problems. Same underlying structure.

RELATIONSHIPS AS FIRST-CLASS CITIZENS

FINAL CLOSE

Thank you.

QUESTIONS & DISCUSSION